

Polynomials via completions of parametrized categories

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My motivation

- ▶ The introduction to David Chan's paper
[*Bi-incomplete Tambara functors as $\mathcal{O}$ -commutative monoids*,
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provides good background to the roles of Mackey and Tambara
functors in equivariant stable homotopy theory.
- ▶ In particular, the third paragraph begins with the following sentence:
*While Mackey functors should be thought of as the correct analogue
of an abelian group in the equivariant setting, the role of commutative
rings is played by objects known as Tambara functors.*
- ▶ Why? I wanted to understand this better by approaching it at the
objective level.
- ▶ Distributive laws come to mind.
- ▶ Let's see!

Parametrized categories

- ▶ Categories varying over a category or bicategory $\mathcal{C}$ can be regarded as pseudofunctors $\mathbb{X} : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$.
- ▶ We think of an object of $\mathbb{X}U$ as a *generalised object* of the $\mathcal{C}$ -variable category $\mathbb{X}$.
- ▶ We write

$$\text{CAT}\mathcal{C} = \text{Ps}(\mathcal{C}^{\text{op}}, \text{Cat})$$

for the (strict) bicategory of pseudofunctors (also called “homomorphisms”), pseudonatural transformations, and modifications.

- ▶ A $\mathcal{C}$ -variable category $\mathbb{X} \in \text{CAT}\mathcal{C}$ is *cocomplete with respect to a morphism* $j : \mathbb{A} \rightarrow \mathbb{B}$ in $\text{CAT}\mathcal{C}$ [17] when every morphism $f : \mathbb{A} \rightarrow \mathbb{X}$ has a pointwise [8] left extension along j .
- ▶ Generally, pointwiseness translates to a lax form of the Chevalley-Beck (CB) condition. The pseudo form is relevant when dealing with groupoid fibrations as we will now remind you.

Groupoid fibrations

- ▶ A morphism $x : Z \rightarrow X$ in $\mathcal{E}$ is *cartesian* for a functor $P : \mathcal{E} \rightarrow \mathcal{B}$ when (like initial topology on Z when $\mathcal{E} = \mathbf{Top}$ and $\mathcal{B} = \mathbf{Set}$)

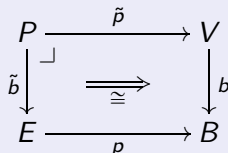
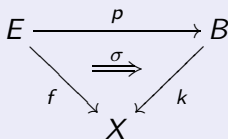
$$\begin{array}{ccc} \mathcal{E}(K, Z) & \xrightarrow{\mathcal{E}(K, x)} & \mathcal{E}(K, X) \\ P \downarrow & & \downarrow P \\ \mathcal{B}(PK, PZ) & \xrightarrow{\mathcal{B}(PK, Px)} & \mathcal{B}(PK, PX) \end{array}$$

is a pullback in $\mathbf{Set}$ for all objects K of $\mathcal{E}$.

- ▶ P is a *groupoid fibration (over $\mathcal{B}$)* when every morphism of $\mathcal{E}$ is cartesian for P and, for all $X \in \mathcal{E}$ and $B \xrightarrow{f} PX$ in $\mathcal{B}$, there exists a morphism $Z \xrightarrow{x} X$ in $\mathcal{E}$ and an isomorphism $B \cong PZ$ whose composite with Px is f .
- ▶ P is a *groupoid opfibration* when $\mathcal{E}^{\text{op}} \xrightarrow{P} \mathcal{B}^{\text{op}}$ is a groupoid fibration.
- ▶ A morphism $p : E \rightarrow B$ in a bicategory $\mathcal{C}$ is a *groupoid fibration* when each functor $\mathcal{C}(U, p) : \mathcal{C}(U, E) \rightarrow \mathcal{C}(U, B)$ is a groupoid fibration.

Proposition

In any finitely complete bicategory, suppose $p : E \rightarrow B$ is a groupoid opfibration. In diagram (1), the 2-morphism σ exhibits k as a pointwise left extension of f along p if and only if, for all morphisms b with the square a bicategorical pullback (briefly, bipullback), the pasted composite at p exhibits kb as a left extension of $f\tilde{b}$ along $\tilde{p}$.



(1)

$\mathcal{R}$ -Cocompleteness

- ▶ If $\mathcal{R}$ is a set of morphisms of a bicategory $\mathcal{C}$, we say $\mathbb{X}$ is $\mathcal{R}$ -cocomplete when it is cocomplete with respect to all morphisms $\mathcal{C}(-, j) : \mathcal{C}(-, U) \rightarrow \mathcal{C}(-, V)$ for $j : U \rightarrow V$ in $\mathcal{R}$.
- ▶ A pseudonatural transformation $\theta : \mathbb{X} \rightarrow \mathbb{Y}$ is $\mathcal{R}$ -cocontinuous when it preserves left extensions along the $\mathcal{C}(-, j)$ for $j \in \mathcal{R}$.
- ▶ Let $\text{CAT}_{\mathcal{R}\mathcal{C}}$ denote the subcategory of $\text{CAT}_{\mathcal{C}}$ consisting of the $\mathcal{R}$ -cocomplete objects and $\mathcal{R}$ -cocontinuous morphisms.

Definition

A *protocalibration* of a bicategory $\mathcal{C}$ is a set $\mathcal{R}$ of morphisms when it satisfies the four conditions:

- I. all equivalences are in $\mathcal{R}$ and any morphism isomorphic to one in $\mathcal{R}$ is in $\mathcal{R}$;
- C. composites of morphisms in $\mathcal{R}$ are in $\mathcal{R}$;
- P. for all $U \xrightarrow{f} W$ in $\mathcal{R}$ and all $V \xrightarrow{g} W$ in $\mathcal{C}$, the bicategorical pullback $P \xrightarrow{\bar{f}} V$ of f along g exists and is in $\mathcal{R}$;
- G. every morphism in $\mathcal{R}$ is a groupoid opfibration.

Each protocalibration $\mathcal{R}$ will be identified with the subcategory of $\mathcal{C}$ containing all objects, only the morphisms in $\mathcal{R}$, and all 2-morphisms between them. Notice that a protocalibration of $\mathcal{C}^{\text{co}}$ is the same as one for $\mathcal{C}$ except for condition (G) where “groupoid opfibration” must be replaced by “groupoid fibration”.

Example

1. Suppose $\mathcal{C}$ is just a category (locally discrete as a bicategory). Then every morphism is both a groupoid fibration and a groupoid opfibration, and bicategorical pullbacks are pullbacks. If $\mathcal{C}$ has pullbacks, the set of all morphisms is a protocalibration.
2. Again suppose $\mathcal{C}$ is a category. If pullbacks of powerful (= exponentiable) morphisms exist then the powerful morphisms form a protocalibration (see Corollary 2.6 of [104]).
3. Let $\mathcal{C}$ be a category with pullbacks and every morphism powerful. The groupoid fibrations (= left-adjoint morphisms) in the bicategory $\mathrm{Spn}\mathcal{C}$ of spans in $\mathcal{C}$ is a protocalibration of $(\mathrm{Spn}\mathcal{C})^{\mathrm{co}}$ (see Propositions 4.1 and 8.3 of [134]).

Proposition

Let $\mathcal{R}$ be a protocalibration of the bicategory $\mathcal{C}$. A $\mathcal{C}$ -variable category $\mathbb{X}$ is $\mathcal{R}$ -cocomplete if and only if, for all $f \in \mathcal{R}$, the functor $\mathbb{X}f$ has a left adjoint $\mathbb{X}_*f$ and, for all bicategorical pullback squares

$$\begin{array}{ccc}
 P & \xrightarrow{f_g} & V \\
 g_f \downarrow \lrcorner & \xRightarrow{\cong} & \downarrow g \\
 U & \xrightarrow{f} & W
 \end{array} \tag{2}$$

in $\mathcal{C}$, the mate $\mathbb{X}_*f_g \circ \mathbb{X}g_f \Rightarrow \mathbb{X}g \circ \mathbb{X}_*f$ of the isomorphism $\mathbb{X}g_f \circ \mathbb{X}f \cong \mathbb{X}f_g \circ \mathbb{X}g$ is invertible. A pseudonatural transformation $\theta : \mathbb{X} \rightarrow \mathbb{Y}$ is $\mathcal{R}$ -cocontinuous if and only if, for all $U \xrightarrow{f} W$ in $\mathcal{R}$, the mate $\theta_f : \mathbb{Y}_*f \circ \theta_U \rightarrow \theta_W \circ \mathbb{X}_*f$ of $\theta_f : \theta_U \circ \mathbb{X}f \rightarrow \mathbb{Y}f \circ \theta_W$ is invertible.

See [13] Proposition 9.12 and [17] Proposition 7.12.

Lemma

Suppose $\mathcal{C}$ has binary bicategorical products¹ and $\mathcal{R}$ is a protocalibration of $\mathcal{C}$. If $X \xrightarrow{f} Y$ and $X' \xrightarrow{f'} Y'$ are in $\mathcal{R}$ then so too is $X \times X' \xrightarrow{f \times f'} Y \times Y'$. So, if $\mathcal{C}$ has finite bicategorical products then $\mathcal{R}$ is a monoidal subbcategory of the cartesian monoidal bicategory $\mathcal{C}$.

Proof.

The morphism $X \times X' \xrightarrow{f \times f'} Y \times Y'$ is the composite of $f \times 1_{X'}$ and $1_Y \times f'$ which are respectively bipullbacks of f and f' along projections. $\square$

¹We will abbreviate this to “biproduct” and use “direct sum” when product and coproduct coincide.

Corollary

Suppose $\mathcal{C}$ has binary biproducts and $\mathcal{R}$ is a protocalibration of $\mathcal{C}$. If $\mathbb{Z}$ is $\mathcal{R}$ -cocomplete then so is $\mathbb{Z}(- \times W)$ for all $W \in \mathcal{C}$.

Proof.

Cocompleteness of $\mathbb{Z}(- \times W)$ with respect to any f amounts, by Yoneda, to cocompleteness of $\mathbb{Z}$ with respect to $f \times 1_W$. $\square$

Proposition

Suppose $\mathcal{C}$ has binary biproducts and $\mathcal{R}$ is a protocalibration of $\mathcal{C}$. For $\mathbb{Y}$ and $\mathbb{Z}$ in $\text{CAT}_{\mathcal{R}}\mathcal{C}$, the object $[\mathbb{Y}, \mathbb{Z}]_{\mathcal{R}} \in \text{CAT}\mathcal{C}$, defined on objects by

$$[\mathbb{Y}, \mathbb{Z}]_{\mathcal{R}} W = \text{CAT}_{\mathcal{R}}\mathcal{C}(\mathbb{Y}, \mathbb{Z}(- \times W))$$

and on morphisms $W' \xrightarrow{h} W$ by composition with $\mathbb{Z}(- \times h)$, is in $\text{CAT}_{\mathcal{R}}\mathcal{C}$.

Proof.

We need to see that, if $\theta : \mathbb{Y} \rightarrow \mathbb{Z}(- \times W)$ is $\mathcal{R}$ -cocontinuous, so too is $\mathbb{Z}(- \times h) \circ \theta$. Take $V \xrightarrow{r} U$ in $\mathcal{R}$. The right square in the diagram

$$\begin{array}{ccccc}
 \mathbb{Y}U & \xrightarrow{\theta_U} & \mathbb{Z}(U \times W) & \xrightarrow{\mathbb{Z}(1_U \times h)} & \mathbb{Z}(U \times W') \\
 \mathbb{Y}r \downarrow & \xRightarrow[\cong]{\theta_r} & \downarrow \mathbb{Z}(r \times 1_W) & \xRightarrow[\cong]{} & \downarrow \mathbb{Z}(r \times 1_{W'}) \\
 \mathbb{Y}V & \xrightarrow{\theta_V} & \mathbb{Z}(V \times W) & \xrightarrow{\mathbb{Z}(1_V \times h)} & \mathbb{Z}(V \times W')
 \end{array}$$

is the value of $\mathbb{Z}$ on a bipullback in $\mathcal{C}$. The invertibility of the mate of the pasted 2-morphism follows from $\mathcal{R}$ -cocontinuity of θ and $\mathcal{R}$ -cocompleteness of $\mathbb{Z}$. □

$\mathcal{R}$ -cocompletion

Suppose $\mathcal{R}$ is a protocolibration of the bicategory $\mathcal{C}$. The $\mathcal{R}$ -cocompletion $\overrightarrow{\mathbb{X}}$ of $\mathbb{X} \in \text{CAT}\mathcal{C}$ is described as follows. An object of the category $\overrightarrow{\mathbb{X}} U$ is a pair $(U \xleftarrow{u} S, x \in \mathbb{X}S)$ with $u \in \mathcal{R}$; we might think of such an object as a U -indexed family of generalised objects of $\mathbb{X}$. A morphism

$$(U \xleftarrow{u} S, x \in \mathbb{X}S) \rightarrow (U \xleftarrow{u'} S', x' \in \mathbb{X}S')$$

of $\overrightarrow{\mathbb{X}} U$ is an isomorphism class of triples $(S \xrightarrow{w} S', u'w \cong u, x \xrightarrow{\xi} (\mathbb{X}w)x')$ where ξ is a morphism of $\mathbb{X}S$; if we identify representables with their representing objects, the picture of such a morphism is a lax morphism (3) of spans in $\text{CAT}\mathcal{C}$.

$$\begin{array}{ccccc}
 & & S & & \\
 & \swarrow^{u \in \mathcal{R}} & \downarrow w & \searrow^x & \\
 U & \xleftarrow{u' \in \mathcal{R}} & S' & \xrightarrow{x'} & \mathbb{X}
 \end{array}
 \quad \begin{array}{c} \cong \\ \xi \end{array}
 \quad (3)$$

$\mathcal{R}$ -cocompletion: continued

For $r : V \rightarrow U$ in $\mathcal{C}$, define the functor $\overrightarrow{\mathbb{X}}r : \overrightarrow{\mathbb{X}}U \rightarrow \overrightarrow{\mathbb{X}}V$ to take (u, x) to $(u_r, (\mathbb{X}r_u)x)$ where

$$\begin{array}{ccc}
 P & \xrightarrow{u_r} & V \\
 r_u \downarrow \lrcorner & & \downarrow r \\
 S & \xrightarrow{u} & U
 \end{array}
 \qquad
 \begin{array}{ccc}
 P' & \xrightarrow{u'_r} & V \\
 r_{u'} \downarrow \lrcorner & & \downarrow r \\
 S' & \xrightarrow{u'} & U
 \end{array}
 \tag{4}$$

are bipullbacks, and (w, ξ) to $(\bar{w}, \bar{\xi})$ where $\bar{w}$ is defined by compatible $u'_r \bar{w} \cong u_r$, $r_{u'} \bar{w} \cong wr_u$, and $\bar{\xi}$ by

$$\bar{\xi} := \left((\mathbb{X}r_u)x \xrightarrow{(\mathbb{X}r_u)\xi} (\mathbb{X}r_u)(\mathbb{X}w)x' \cong (\mathbb{X}\bar{w})(\mathbb{X}r_{u'})x' \right) .$$

$\mathcal{R}$ -cocompletion: continued: continued

For a 2-morphism $\alpha : r \Rightarrow s : V \rightarrow U$, let $S \xleftarrow{s_u} Q \xrightarrow{u_s} V$ be the bipullback of the cospan $S \xrightarrow{u} U \xleftarrow{s} V$. Since u is a groupoid opfibration, there exists a cocartesian 2-morphism $\kappa : r_u \Rightarrow k$ and isomorphism $su_r \cong uk$ such that $u\kappa$ is the composite $ur_u \cong ru_r \xrightarrow{\alpha u_r} su_r \cong uk$. If $r \in \mathcal{R}$ then $\overrightarrow{\mathbb{X}}r$ has a left adjoint $\overrightarrow{\mathbb{X}}_*r : \overrightarrow{\mathbb{X}}V \rightarrow \overrightarrow{\mathbb{X}}U$ taking $(V \xleftarrow{v} T, y \in \mathbb{X}T)$ to $(U \xleftarrow{rv} T, y \in \mathbb{X}T)$ (noting that $rv \in \mathcal{R}$ by condition (C)). To verify the pseudo-Chevalley-Beck condition, we refer to the bipullbacks (5) with $f, u \in \mathcal{R}$.

$$\begin{array}{ccccc}
 Q & \xrightarrow{u_{gf}} & P & \xrightarrow{f_g} & V \\
 \downarrow g_{fu} \lrcorner & & \downarrow g_f \lrcorner & & \downarrow g \\
 S & \xrightarrow{u \in \mathcal{R}} & U & \xrightarrow{f \in \mathcal{R}} & W
 \end{array} \tag{5}$$

$$\begin{aligned}
 (\overrightarrow{\mathbb{X}}g \circ \overrightarrow{\mathbb{X}}_*f)(u, x) &= (\overrightarrow{\mathbb{X}}g)(fu, x) \cong (f_g u_{gf}, (\mathbb{X}g_{fu})x) \\
 &= (\overrightarrow{\mathbb{X}}_*f_g)(u_{gf}, (\mathbb{X}g_{fu})x) \cong (\overrightarrow{\mathbb{X}}_*f_g \circ \mathbb{X}g_f)(u, x)
 \end{aligned}$$

naturally in $(u, x) \in \overrightarrow{\mathbb{X}}U$. So $\overrightarrow{\mathbb{X}} \in \text{CAT}\mathcal{E}$ is $\mathcal{R}$ -cocomplete.

A side comment

- ▶ Suppose $\mathcal{C}$ is a category with pullbacks so that we can take $\mathcal{R}$ to contain all morphisms.
- ▶ The terminal object $\mathbf{1}$ of $\text{CAT}\mathcal{C}$ takes all objects of $\mathcal{C}$ to the terminal category. Then $\mathbb{C} := \overrightarrow{\mathbf{1}}$ takes $U \in \mathcal{C}$ to the slice category $\mathcal{C}/U$ and is defined on morphisms using pullback.
- ▶ Notice that an ordinary category has coproducts if and only if all pointwise left Kan extensions into it exist along functors between discrete categories.
- ▶ Also Set is the coproduct cocompletion of the terminal ordinary category.
- ▶ Now we see that $\mathbb{C}$ is the coproduct cocompletion of the terminal $\mathcal{C}$ -variable category.
- ▶ This is one justification for viewing $\mathbb{C}$ as playing the role in $\mathcal{C}$ -variable category theory of Set in ordinary category theory.
- ▶ For a general protocalibration $\mathcal{R}$, we write $\mathbb{C}_{\mathcal{R}}$ for the $\mathcal{R}$ -cocompletion $\overrightarrow{\mathbf{1}}$ of $\mathbf{1}$.

There is a pseudonatural transformation $\eta_{\mathbb{X}} : \mathbb{X} \rightarrow \overrightarrow{\mathbb{X}}$ whose component $\eta_{\mathbb{X}U} : \mathbb{X}U \rightarrow \overrightarrow{\mathbb{X}}U$ at $U \in \mathcal{C}$ is defined by $\eta_{\mathbb{X}U}x = (U \xleftarrow{1_U} U, x \in \mathbb{X}U)$.

Proposition

Suppose $\mathcal{R}$ is a protocolibration of the bicategory $\mathcal{C}$. Then $\mathbb{X} \in \text{CAT}\mathcal{C}$ is $\mathcal{R}$ -cocomplete if and only if $\eta_{\mathbb{X}}$ has a left adjoint in $\text{CAT}\mathcal{C}$.

Proof.

A left adjoint to $\eta_{\mathbb{X}U} : \mathbb{X}U \rightarrow \overrightarrow{\mathbb{X}}U$ precisely gives, for each $(u, x) \in \overrightarrow{\mathbb{X}}U$, an object $u_*(x) \in \mathbb{X}U$ and an isomorphism $\mathbb{X}U(u_*(x), y) \cong \overrightarrow{\mathbb{X}}U((u, x), (1_U, y))$ natural in $y \in \mathbb{X}U$. However, $\overrightarrow{\mathbb{X}}U((u, x), (1_U, y)) \cong \mathbb{X}S(x, (\mathbb{X}u)y)$. So this amounts precisely to having a left adjoint $\mathbb{X}_*u$ to $\mathbb{X}u$ for all $u \in \mathcal{R}$. Pseudonaturality of $\eta_{\mathbb{X}}$ amounts to the $\mathcal{R}$ -Chevalley-Beck condition for $\mathbb{X}$. □

We think of the left adjoint to $\eta_{\mathbb{X}}$ as assigning a coproduct to the indexed families of generalised objects.

Corollary

The pseudonatural transformation $\eta_{\overrightarrow{\mathbb{X}}} : \overrightarrow{\mathbb{X}} \rightarrow \overrightarrow{\overrightarrow{\mathbb{X}}}$ has a left adjoint $\mu_{\mathbb{X}} : \overrightarrow{\overrightarrow{\mathbb{X}}} \rightarrow \overrightarrow{\mathbb{X}}$ in $\text{CAT}\mathcal{C}$.

For each protocalibration $\mathcal{R}$ of $\mathcal{C}$, we have a Kock-Zöberlein pseudomonad

$$\overrightarrow{(-)}_{\mathcal{R}} = ((\overrightarrow{-}), \eta, \mu)$$

on $\text{CAT}\mathcal{C}$.

Corollary

The inclusion of the subcategory $\text{CAT}_{\mathcal{R}}\mathcal{C}$ of $\text{CAT}\mathcal{C}$, consisting of the $\mathcal{R}$ -cocomplete objects and $\mathcal{R}$ -cocontinuous morphisms, is pseudomonadic. That is, the inclusion is biequivalent over $\text{CAT}\mathcal{C}$ to the forgetful pseudofunctor from the bicategory of Eilenberg-Moore pseudo-algebras for the pseudomonad $\overrightarrow{(-)}_{\mathcal{R}}$. In particular, $\text{CAT}_{\mathcal{R}}\mathcal{C}$ is a complete bicategory and bicategorical limits are preserved by the inclusion.

$\mathcal{L}$ -completion

The *opposite* $\mathbb{X}^{\text{op}}$ of $\mathbb{X} \in \text{CAT}\mathcal{C}$ is obtained by composition with the 2-isomorphism $(-)^{\text{op}} : \text{Cat} \rightarrow \text{Cat}^{\text{co}}$. This is the value on objects for a 2-isomorphism

$$(-)^{\text{op}} : \text{CAT}\mathcal{C} \rightarrow (\text{CAT}\mathcal{C}^{\text{co}})^{\text{co}} . \quad (6)$$

Suppose $\mathcal{L}$ is a protocolibration of $\mathcal{C}^{\text{co}}$. We define $\mathbb{X} \in \text{CAT}\mathcal{C}$ to be $\mathcal{L}$ -complete when $\mathbb{X}^{\text{op}}$ is $\mathcal{L}$ -cocomplete. It follows that the $\mathcal{L}$ -completion $\overleftarrow{\mathbb{X}}$ of $\mathbb{X} \in \text{CAT}\mathcal{C}$ is $(\overrightarrow{\mathbb{X}^{\text{op}}})^{\text{op}}$. If $\mathbb{X}$ is $\mathcal{L}$ -complete and $V \xrightarrow{f} U$ is in $\mathcal{L}$, we write $\mathbb{X}_! f$ for the right adjoint of $\mathbb{X}f : \mathbb{X}U \rightarrow \mathbb{X}V$.

We write $_{\mathcal{L}}\text{CAT}\mathcal{C}$ for the locally full subbcategory of $\text{CAT}\mathcal{C}$ consisting of the $\mathcal{L}$ -complete objects and $\mathcal{L}$ -continuous morphisms. (6) induces

$$_{\mathcal{L}}\text{CAT}\mathcal{C} \cong (\text{CAT}_{\mathcal{L}}\mathcal{C}^{\text{co}})^{\text{co}} . \quad (7)$$

We have a dual Kock-Zöberlein pseudomonad on $\text{CAT}\mathcal{C}$ for which the Eilenberg-Moore pseudo-algebras are the $\mathcal{L}$ -complete objects where the algebra action on the object is right adjoint to the unit.

$\mathcal{L}$ -completion: continued

An object of $\overleftarrow{\mathbb{X}}U$ is a pair $(U \xleftarrow{u} S, x \in \mathbb{X}S)$ with $u \in \mathcal{L}$, as for $\overrightarrow{\mathbb{X}}U$. A morphism

$$(U \xleftarrow{u} S, x \in \mathbb{X}S) \rightarrow (U \xleftarrow{u'} S', x' \in \mathbb{X}S')$$

of $\overleftarrow{\mathbb{X}}U$ is an isomorphism class of triples $(S' \xrightarrow{w} S, u' \cong uw, (\mathbb{X}w)x \xrightarrow{\zeta} x')$ where ζ is a morphism of $\mathbb{X}S'$.

$$\begin{array}{ccccc}
 U & \xleftarrow{u \in \mathcal{L}} & S & \xrightarrow{x} & \mathbb{X} \\
 & \searrow^{u' \in \mathcal{L}} & \uparrow w & \nearrow \zeta & \\
 & & S' & \xrightarrow{x'} &
 \end{array}
 \quad (8)$$

For $r : V \rightarrow U$ in $\mathcal{E}$, the functor $\overleftarrow{\mathbb{X}}r : \overleftarrow{\mathbb{X}}U \rightarrow \overleftarrow{\mathbb{X}}V$ takes the object (u, x) to $(u_r, (\mathbb{X}r_u)x)$ with notation as in (4) which is used also in defining $\overleftarrow{\mathbb{X}}r$ on morphisms. The unit $\rho_{\mathbb{X}} : \mathbb{X} \rightarrow \overleftarrow{\mathbb{X}}$ for the pseudomonad has components at U the functor $\rho_U : (x \in \mathbb{X}U) \mapsto ((1_U, x) \in \overleftarrow{\mathbb{X}}U)$ and $(x \xrightarrow{\zeta} x') \mapsto (1_U, \zeta)$.

$\mathcal{L}$ -completion: continued: continued

Corollary

Suppose $\mathcal{L}$ is a protocalibration of the bicategory $\mathcal{C}^{\text{co}}$. Then $\mathbb{X} \in \text{CAT}\mathcal{C}$ is $\mathcal{L}$ -complete if and only if $\rho_{\mathbb{X}}$ has a right adjoint in $\text{CAT}\mathcal{C}$.

We think of the right adjoint to $\rho_{\mathbb{X}}$ as assigning a product to the indexed families of generalised objects of $\mathbb{X}$.

$\mathcal{R}$ -Spans and $\mathcal{R}$ -cocompleteness

We look at slight variants of Bénabou's bicategory of spans. Let $\mathcal{R}$ be a protocalibration of the bicategory $\mathcal{C}$. A (left) $\mathcal{R}$ -span from U to V in $\mathcal{C}$ is a diagram $U \xleftarrow{u} S \xrightarrow{v} V$ in $\mathcal{C}$ with $u \in \mathcal{R}$. There is a bicategory $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}$ whose objects are those of $\mathcal{C}$, whose morphisms are the $\mathcal{R}$ -spans, and whose 2-morphisms (denoted $[f, \sigma]$ by abuse of language) are the isomorphism classes of diagrams:

$$\begin{array}{ccccc}
 & & S & & \\
 & \swarrow^{u \in \mathcal{R}} & \downarrow f & \searrow^v & \\
 U & \xleftarrow{u' \in \mathcal{R}} & S' & \xrightarrow{v'} & V
 \end{array}
 \quad \begin{array}{c} \Leftarrow \sigma \Leftarrow \end{array}
 \quad (9)$$

The composition of $\mathcal{R}$ -spans is defined on morphisms using bipullback. To define the composition on 2-morphisms, we use the constructions (in dual form for groupoid opfibrations rather than fibrations) given in Proposition 5.2 (iii) and Proposition 8.4 of [134].

We have a pseudofunctor

$$(-)_* : \mathcal{C} \rightarrow \mathrm{Spn}_{\mathcal{R}} \mathcal{C} \quad (10)$$

taking the morphism $U \xrightarrow{r} V$ to the $\mathcal{R}$ -span $r_* = (U \xleftarrow{1_U} U \xrightarrow{r} V)$ and the 2-morphism $\alpha : r \Rightarrow s$ to the 2-morphism $[1_U, \alpha]$. If $r \in \mathcal{R}$ then r_* has a right adjoint $r^* = (V \xleftarrow{r} U \xrightarrow{1_U} U)$ in $\mathrm{Spn}_{\mathcal{R}} \mathcal{C}$. Every morphism $U \xleftarrow{u} S \xrightarrow{v} V$ of $\mathrm{Spn}_{\mathcal{R}} \mathcal{C}$ decomposes as $v_* \circ u^*$.

Restriction along the opposite of (10) defines a pseudomonadic pseudofunctor

$$\mathrm{CATSpn}_{\mathcal{R}} \mathcal{C} \rightarrow \mathrm{CAT} \mathcal{C} . \quad (11)$$

Re-identification of the pseudo-algebras as follows is an objective variant of Lindner's result of 1976.

Proposition

Restriction along the opposite of (10) defines a bicategorical equivalence

$$\mathrm{CATSpn}_{\mathcal{R}} \mathcal{C} \sim \mathrm{CAT} \mathcal{C} .$$

Let $\mathcal{L}$ be a protocalibration of the bicategory $\mathcal{C}^{\text{co}}$. So we can define

$${}_{\mathcal{L}}\text{Spn}\mathcal{C} = (\text{Spn}_{{}_{\mathcal{L}}}\mathcal{C}^{\text{co}})^{\text{co}}.$$

The 2-morphisms are represented by diagrams of the form (12).

$$\begin{array}{ccccc}
 & & S & & \\
 & \swarrow^{u \in \mathcal{L}} & \uparrow f & \searrow v & \\
 U & & S' & & V \\
 & \swarrow_{u' \in \mathcal{L}} & \downarrow & \searrow_{v'} & \\
 & & & &
 \end{array}
 \begin{array}{c}
 \cong \\
 \cong
 \end{array}
 \quad (12)$$

Proposition

Restriction along the opposite of $(-)_ : \mathcal{C} \rightarrow {}_{\mathcal{L}}\text{Spn}\mathcal{C}$ defines a bicategorical equivalence*

$$\text{CAT}({}_{\mathcal{L}}\text{Spn}\mathcal{C}) \sim {}_{\mathcal{L}}\text{CAT}\mathcal{C}.$$

Some terminology and notation

Let $\mathcal{C}$ be a category with pullbacks. The functor defined by pulling back along $u : S \rightarrow U$ is denoted by $\Delta_u : \mathcal{C}/U \rightarrow \mathcal{C}/S$. There is always a left adjoint to Δ_u denoted by Σ_u and obtained by composition with u . Recall that a morphism $u : S \rightarrow U$ is called *powerful* (or “exponentiable”) when Δ_u has a right adjoint Π_u .

Let $\mathcal{P}$ denote the protocalibration of powerful morphisms in $\mathcal{C}$.

We are particularly interested in the $\mathcal{C}$ -cocompletion and the $\mathcal{P}$ -completion.

So, for now, by $\overrightarrow{\mathbb{X}}$ we will mean the $\mathcal{C}$ -cocompletion of $\mathbb{X} \in \text{CAT}\mathcal{C}$ and by $\overleftarrow{\mathbb{X}}$ we will mean the $\mathcal{P}$ -completion of $\mathbb{X}$.

We write $\text{CAT}_{\Sigma}\mathcal{C}$ for $\text{CAT}_{\mathcal{C}}\mathcal{C}$ and ${}_{\Pi}\text{CAT}\mathcal{C}$ for ${}_{\mathcal{P}}\text{CAT}\mathcal{C}$.

A distributive law

There is a bicategorical distributive law

$$\lambda : \overleftarrow{(-)} \circ \overrightarrow{(-)} \longrightarrow \overrightarrow{(-)} \circ \overleftarrow{(-)}$$

where the component functor

$$\lambda_U : (\overleftrightarrow{\mathbb{X}})U \longrightarrow \overrightarrow{(\mathbb{X})}U$$

takes $(U \xleftarrow{u} S, S \xleftarrow{a} A, x \in \mathbb{X}A)$, where $u \in \mathcal{P}$, to

$$(U \xleftarrow{\Pi_u a} B, B \xleftarrow{\text{proj}_2} S \times_U B, (\mathbb{X}e)x \in \mathbb{X}(S \times_U B))$$

where e comes from the counit of $\Delta_u \dashv \Pi_u$.

Proposition

If $\mathbb{X} \in \text{CAT}\mathcal{C}$ is $\mathcal{P}$ -complete then so is the $\mathcal{C}$ -cocompletion $\overleftrightarrow{\mathbb{X}}$. Moreover, the unit $\eta_{\mathbb{X}} : \mathbb{X} \rightarrow \overleftrightarrow{\mathbb{X}}$ is in $\prod \text{CAT}\mathcal{C}$.

The pseudomonad $\overrightarrow{(-)}$ on $\text{CAT}\mathcal{C}$ therefore lifts to $\text{CAT}_{\Pi}\mathcal{C}$.

$$\begin{array}{ccc}
 \Pi\text{CAT}\mathcal{C} & \xrightarrow{\overrightarrow{(-)}_{\Pi}} & \Pi\text{CAT}\mathcal{C} \\
 \text{incl} \downarrow & \xRightarrow{\simeq} & \downarrow \text{incl} \\
 \text{CAT}\mathcal{C} & \xrightarrow{\overrightarrow{(-)}} & \text{CAT}\mathcal{C}
 \end{array} \tag{13}$$

The pseudo-algebras for the lifted pseudomonad are those of the composite pseudomonad $\mathbb{X} \mapsto \overleftrightarrow{(\mathbb{X})}$ on $\text{CAT}\mathcal{C}$ and form the 2-category at the top of the diamond

$$\begin{array}{ccc}
 & \Pi\text{CAT}_{\Sigma}\mathcal{C} & \\
 \swarrow & & \searrow \\
 \Pi\text{CAT}\mathcal{C} & & \text{CAT}_{\Sigma}\mathcal{C} \\
 \searrow & & \swarrow \\
 & \text{CAT}\mathcal{C} &
 \end{array} \tag{14}$$

The bicategory of polynomials

Let $\mathcal{E}$ be a category with pullbacks. There is a bicategory $\text{Ply}\mathcal{E}$ of polynomials in $\mathcal{E}$. The objects are those of $\mathcal{E}$. The morphisms from X to Y are *polynomials* $p = (X \xleftarrow{r} A \xrightarrow{n} B \xrightarrow{t} Y)$ in $\mathcal{E}$ with n powerful. The morphisms between polynomials are isomorphism classes of commutative diagrams

$$\begin{array}{ccccc}
 & A & \xrightarrow{n} & B & \\
 & \uparrow \ell & & \uparrow 1_B & \\
 X & \xleftarrow{r} & A & \xrightarrow{n} & B & \xrightarrow{t} & Y \\
 & \downarrow g_{n'} & & \downarrow g & & \\
 & A' & \xrightarrow{n'} & B' & \\
 & \uparrow r' & & \uparrow t' &
 \end{array}$$

(15)

Composition uses bipullback of spans of spans.

Polynomials as spans of spans

Recall the philosophy of [134] that it is useful to regard p as a span

$$p = (X \xleftarrow{(n,A,r)} B \xrightarrow{(1_B,B,t)} Y) \quad (16)$$

in $\mathbf{Spn}\mathcal{E}$. Then the morphisms interpret as diagrams

$$\begin{array}{ccccc} & & B & & \\ & \swarrow (n,A,r) & \downarrow \lambda(1,B,g) = & \searrow (1,B,t) & \\ X & \xleftarrow{(n',A',r')} & B' & \xrightarrow{(1,B',t')} & Y \end{array} \quad (17)$$

Powerful morphisms form a protocalibration $\mathcal{P}$ of $\mathcal{E}$ so we have the bicategory $\mathcal{P}\mathbf{Spn}\mathcal{E}$.

Lemma

The groupoid fibrations in the bicategory $\mathbf{Spn}\mathcal{E}$ are morphisms in $(\mathcal{P}\mathbf{Spn}\mathcal{E})^{\mathrm{co}}$ and, as such form a protocalibration denoted by LS.

Polynomials as spans of spans: continued

Theorem

If $\mathcal{E}$ is a category with pullbacks then

$$(\mathrm{Ply}\mathcal{E})^{\mathrm{op}} = \mathrm{Spn}_{\mathrm{LS}}(\mathcal{P}\mathrm{Spn}\mathcal{E}) .$$

Proof.

This is an unpacking exercise, just note that the 2-morphism λ in $\mathcal{P}\mathrm{Spn}\mathcal{E}$ is in the reverse direction as a morphism of spans. $\square$

A view of Charles Walker's result in JPAA **223** (2019)

Corollary

$$\mathrm{Ps}(\mathrm{Ply}\mathcal{E}, \mathrm{Cat}) \sim \coprod \mathrm{CAT}_{\Sigma\mathcal{E}}$$

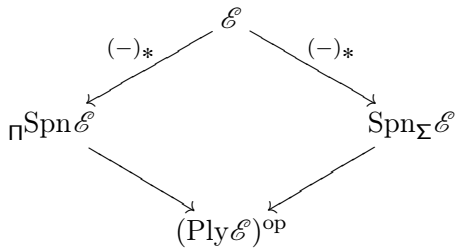
Proof.

Using the Theorem and two earlier Propositions, we have

$$\begin{aligned} \mathrm{Ps}(\mathrm{Ply}\mathcal{E}, \mathrm{Cat}) &\sim \mathrm{CAT}(\mathrm{Spn}_{\mathrm{LS}}(\mathscr{P}\mathrm{Spn}\mathcal{E})) \\ &\sim \mathrm{CAT}_{\mathrm{LS}}(\mathscr{P}\mathrm{Spn}\mathcal{E}) \\ &\sim \mathscr{P}\mathrm{CAT}_{\mathcal{E}\mathcal{E}} = \coprod \mathrm{CAT}_{\Sigma\mathcal{E}} \end{aligned}$$

noting that $(-)_* : \mathcal{E} \rightarrow \mathscr{P}\mathrm{Spn}\mathcal{E}$ maps $\mathcal{E}$ to LS. □

CONCLUSION: $\mathcal{E}$ -variable categories admitting internal products and coproducts can be reconceived as arbitrary $(\mathrm{Ply}\mathcal{E})^{\mathrm{op}}$ -variable categories.



Thank You

